# Gravity above the surface

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use distance from a planet's center to compare fields at altitude.

**SUCCESS CRITERIA - I can...**

1. Use r=R+h rather than altitude alone.
2. Compare fields with the inverse-square relation.
3. Explain why orbit is not a zero-gravity region.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

For fixed masses, what separation multiplier makes gravity **1/9** of its original value? What multiplier is needed if both masses also become **5** times as large?

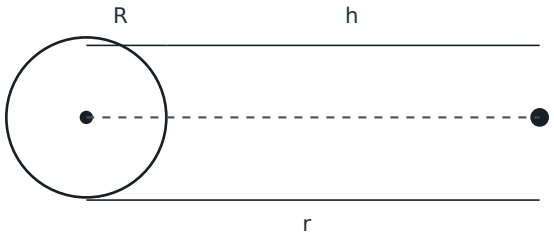
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $r = R + h$; $g_h/g_{surface} = [R/(R + h)]^2$.
R is planet radius; h is altitude. $g_h$ is field strength at altitude h.

**Worked reasoning:** At altitude h=**2** R above a planet's surface, center distance is r=R+h=**3** R. A surface field of **18 N/kg** becomes **18/9=2 N/kg**.

**Your turn:** Draw the radius and altitude as different line segments. **Mark the full center-to-object distance**.

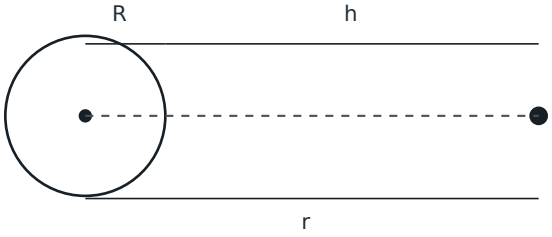



Measure r from the center: r = R + h.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

A satellite is one planet-radius above the surface.
1. **Label R**, h and r on a sketch.
2. What is r in units of R?

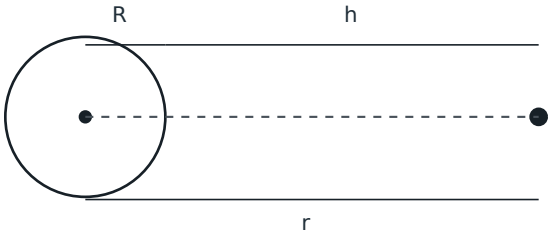



Label surface radius R, altitude h and center distance r.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

For Earth use$R=\mathbf{6.4\times10^6\ m}$, $g_{surface}=\mathbf{9.8\ N/kg}$. **Find g** at $h=\mathbf{4.0\times10^5\ m}$ and at h=R.




**Marzano check - circle:** ① ② ③

# Gravity above the surface

**antocciphysics**



Continue down each column.

## 05 Explain your reasoning

**Explain why an astronaut** at modest orbital altitude can float even though gravity is only slightly weaker there.

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A **4 kg** box has $\mu_s$=**0.50** and $\mu_k$=**0.40** on a level floor; g=**9.8**. **Compare pushes of 8 N** and **24 N**, starting from rest. **Find actual friction** and acceleration.
**Draw the forces** on the chosen object and **label their sources**.

Add and label force arrows.

**Marzano check - circle:** ① ② ③

## 07 Exit check

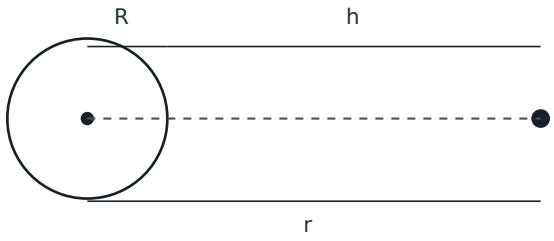
At what altitude in units ofR is field one quarter of surface value? At what altitude is it one ninth?

| Field / surface field | r / R | h / R |
|---|---|---|
| ___ | ___ | ___ |
| ___ | ___ | ___ |

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

1. **Find the altitude h**/R where a spherical planet's field is **10%** of the surface value.
2. **Explain why h**/R is not √**10**.




**Marzano check - circle:** ① ② ③